

How much arithmetic survives a decimal expansion?

Notes accompanying the `ratrecon` library

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Notes on the information content of finite decimal observations of rationals — what can be recovered, at what cost, and what is inevitably lost. These are expository notes: the theorems are classical; the framing, and the way the implementation in `ratrecon` realises each bound, is what the notes are for.

1. Setup

Fix a height bound Q and write $\mathcal{F}_Q = \{p/q \in \mathbb{Q} : 1 \leq q \leq Q\}$ (numerator bounds enter trivially and are ignored here). The *observation map*

$$\rho_k : x \mapsto \text{first } k \text{ decimal digits of } x \text{ (rounded)}$$

sends a rational to a finite string. Two questions:

1. **Exact regime.** When is ρ_k injective on $\mathcal{F}_Q$, and how cheaply can it be inverted?
2. **Lossy regime.** When it is not injective — or when the digits carry noise — what structure remains, and how should candidates be ranked?

Everything in `ratrecon` is an algorithmic shadow of the answers below.

Experimental results quoted below are stored in `results/`; the identifiability benchmark is also reproduced live by `demo.py`, and the learned-prior experiment by `hybrid_gru.py`.

2. The exact regime

2.1 The uniqueness threshold

Theorem 1. *If $10^k > Q^2$ then ρ_k is injective on $\mathcal{F}_Q$: k rounded digits determine p/q with $q \leq Q$ uniquely.*

Proof. Distinct $p/q \neq p'/q'$ with $q, q' \leq Q$ satisfy

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{qq'} \geq \frac{1}{Q^2},$$

since the integer $|pq' - p'q|$ is nonzero. Two reals producing the same k -digit rounding lie in a common interval of length $10^{-k} < 1/Q^2$. ■

The bound is essentially sharp: consecutive Farey fractions of order Q have gaps as small as $\sim 1/Q^2$, so with $10^k \ll Q^2$ collisions are not just possible but generic. `ratrecon.sufficient_digits` computes the threshold with integer arithmetic; the end-to-end tests exercise exactness at the threshold.

2.2 The encoding is near-optimal

$|\mathcal{F}_Q \cap (0, 1]| = \sum_{q \leq Q} \varphi(q) = \frac{3}{\pi^2} Q^2 + O(Q \log Q)$, so identifying a member costs $\log_2(\frac{3}{\pi^2} Q^2) = 2 \log_2 Q - 1.72 + o(1)$ bits, while k digits carry $k \log_2 10$ bits. Theorem 1 needs $k \log_2 10 \geq 2 \log_2 Q$, so the *continuous* separation requirement sits 1.72 bits above the set entropy.

Two distinct overheads are easy to conflate here, so state both: on top of those 1.72 bits, k must be a whole number of digits, which costs up to a further $\log_2 10 \approx 3.32$ bits of granularity — worst case just under five bits total, not two. (At $Q = 100$: entropy ≈ 11.6 bits, continuous requirement 13.3, and $k = 4$ digits carry 13.3 — here the rounding costs nothing, which is exactly the equality case of §7.1.) Digits are still a near-optimal — but *arithmetically scrambled* — encoding of (p, q) ; the entire subject is the cost of unscrambling.

2.3 Inversion in logarithmic time

Brute force over $\mathcal{F}_Q$ costs $\Theta(Q^2)$. Continued fractions invert in $O(k)$ arithmetic steps (on integers of $O(k)$ digits, so $O(k^2)$ bit operations with schoolbook arithmetic):

Theorem 2 (Legendre). *If $|x - p/q| < 1/(2q^2)$ then p/q is a convergent of the continued fraction of x .*

Above the threshold, the true p/q satisfies Legendre’s inequality with respect to the observed value, so it appears among the $O(\log Q)$ convergents of the digit string — this is why `cf_terms` + `convergents` suffices, and why `ratrecon` marks candidates passing the Legendre test as *certified convergents*. Note the precise scope: the criterion certifies membership in the convergent sequence, nothing more — several convergents can pass at once (both $22/7$ and $355/113$ do against π), and uniqueness of a reconstruction is established only by exhaustive bounded enumeration (`identify`).

The interval refinement of this idea is the **Stern–Brocot descent** (`simplest_in_interval`): the observed digits pin x into an interval I , and the smallest-denominator fraction in I is found by descending the Stern–Brocot tree, one continued-fraction term per step (Richards 1981). A useful uniqueness fact justifies “the” simplest fraction: two fractions with the same denominator $q \geq 2$ can never be adjacent in a Farey sequence (adjacency forces $q(c - a) = 1$), so the minimal denominator in an interval is attained exactly once.

3. Below the threshold: the ambiguity set

When $10^k \leq Q^2$ the fibre $\rho_k^{-1}(\text{digits}) \cap \mathcal{F}_Q$ is the set of Farey fractions in an interval I of length 10^{-k} ; its expected size is

$$\mathbb{E} \#(\mathcal{F}_Q \cap I) \approx \frac{3}{\pi^2} Q^2 10^{-k},$$

which the `farey_in_interval` neighbour-walk enumerates at $O(1)$ amortised cost per element. Consistency with the interval is treated as *hard* evidence (zero likelihood outside it); surviving candidates are ranked by the MAP score

$$\log \text{score}(p/q) = -s \ln q - \frac{1}{2} \left(\frac{|x_{\text{obs}} - p/q|}{\sigma \cdot 10^{-k}} \right)^2,$$

with a prior $P(p/q) \propto q^{-s}$ and a Gaussian term in the residual measured in ulps.

What the Gaussian term is, and is not. Under pure deterministic quantisation the likelihood is already exhausted by the cell: it is 1 for every fraction inside and 0 outside, so no residual term follows from the rounding model — inside the cell, all candidates are equally compatible. The Gaussian is therefore a *centre-preference surrogate*: it is motivated by an analogue-noise model (the regime `noise_ulp` is designed for, where σ grows and the cell widens) but is not the exact quantised-data likelihood, which would integrate the noise density over the cell. Stated plainly: **candidates are hard-filtered by the quantisation cell, then optionally ranked by a Gaussian centre preference.** With $s = 2$ and exact digits the residual term is nearly inert anyway — the prior does the discriminating.

On the choice of prior — a point worth being precise about. The default $s = 2$ is a *complexity (Occam) prior*: it charges each candidate its description length $2 \log q$ and makes the simplest consistent fraction win. It is **not** the density of the bounded Farey set: among reduced fractions in $(0, 1]$ with $q \leq Q$ there are $\varphi(q)$ with denominator exactly q , so sampling that set uniformly gives $P(q) \propto \varphi(q)$ — an *increasing* function of q . The two priors give opposite advice, and below the threshold the prior dominates the ranking — so whichever prior better

matches the law generating the truths wins the benchmark, and swapping the source law swaps the winner. Measured top-1 accuracy below the threshold therefore reflects prior/source match, not algorithmic quality, which is why **reconstruct** exposes the exponent s ($s = 0$ recovers maximum-likelihood, closest-to-observation ranking) instead of hard-coding a philosophy. For “naturally occurring” constants — quoted probabilities, tick prices, human-chosen ratios — small denominators really are more probable, and $s = 2$ is the sensible default.

Section 5 of **demo.py** walks the same fraction through $k = 3, \dots, 6$ digits and watches the fibre shrink from 49 candidates to 1 at $k = 5$, one digit before the universal worst-case threshold — illustrating that the bound is sufficient, not necessary.

4. Graph representation: the Stern–Brocot tree

The Stern–Brocot tree is a bijection between positive rationals and finite words in $\{L, R\}^*$; the run-lengths of the word are precisely the continued fraction terms $[a_0; a_1, a_2, \dots]$. This is the natural *graph-based representation* of a rational: a vertex of an infinite binary tree, or equivalently a vertex of the Farey graph (fractions joined when $|pq' - p'q| = 1$).

What does a decimal prefix know about the tree position? The digit interval I selects the set of tree nodes lying in I , and the longest L/R prefix shared by every candidate — the common Stern–Brocot cylinder — is exactly the part of the word that k digits have irrevocably determined. For a closed interval the deepest common ancestor of that set is the simplest fraction in I ; when an endpoint is excluded (as in the half-open observation cells) the ancestor can be the excluded endpoint itself, with the simplest *admissible* fraction sitting just inside at the pruning frontier (**simplest_in_interval**). Digit refinement is monotone tree pruning.

So translating “digit knowledge” into “graph knowledge” preserves a *prefix* of the algebraic data $(a_0, a_1, \dots)$ and destroys the rest. How long is the preserved prefix? That is a theorem:

Theorem 3 (Lochs, 1964). *For almost every x , the number $m(k)$ of continued-fraction terms determined by the first k decimal digits satisfies*

$$\frac{m(k)}{k} \rightarrow \frac{6 \ln 2 \ln 10}{\pi^2} = 0.9702 \dots$$

One decimal digit buys (almost surely, asymptotically) just under one continued-fraction term — the two representations have nearly equal entropy rates, decimal’s being $\ln 10$ per digit and the CF’s being the entropy $\pi^2/(6 \ln 2)$ of the Gauss map per term. The constant is not 1: the representations are *incommensurable*, and the conversion cost is precisely the ratio of the two entropies. For rationals (finite words) the statement degrades gracefully: the prefix determined is governed by the same rate until the expansion terminates.

Worst cases are far from the average: near a high branch of small height (e.g. $x \approx 1/2$), many digits can be spent confirming a single huge CF term a_n — digits determine tree *depth* at rate $\log_2 10$, but CF *terms* only at the Lochs rate, and a single run can hoard the depth budget.

5. Geometric representation: the Farey tessellation

The hyperbolic plane $\mathbb{H}$ carries the Farey tessellation: ideal triangles on the vertices $\mathbb{Q} \cup \{\infty\}$, with p/q and p'/q' joined by a geodesic iff $|pq' - p'q| = 1$ — the Farey graph of §4 made of geodesics. Attach to each p/q its **Ford circle**, the horocycle of Euclidean radius $1/(2q^2)$ tangent at p/q : *the denominator is literally the geometric scale at which the rational is visible*.

A real x becomes the vertical geodesic γ_x ending at x ; the sequence of tessellation edges crossed by γ_x (its *cutting sequence*) is an L/R word whose run-lengths are again the continued-fraction terms (Series 1985). The three representations —

- digit string (analytic),
- Stern–Brocot / Farey-graph path (combinatorial/graph),
- cutting sequence of a geodesic on the modular surface (geometric)

— are three coordinatisations of the same algebraic object, and the information exchange rates between them are governed by one mechanism (the statements below are classical; the point is the unified reading):

- Knowing x to precision ε perturbs the endpoint of γ_x by ε ; the cutting sequence is unchanged until the geodesic enters Ford circles of radius $\lesssim \varepsilon$, i.e. until convergent denominators reach $q_n^2 \gtrsim 1/\varepsilon$. With $\varepsilon = 10^{-k}$ this recovers Theorem 1’s threshold $q^2 < 10^k$, now as a statement about horoball depth.
- Denominator growth is exponential along the geodesic ($q_n^{1/n} \rightarrow e^{\pi^2/(12 \ln 2)}$ a.e., Lévy), which converts the geometric statement into the Lochs rate of §4: depth $\sim$ time $\sim$ digits, terms $\sim$ time $\times$ entropy ratio.

This is the sense in which “translation into graph-based or geometric representations” preserves algebraic information: *everything above scale ε survives every one of these translations; nothing below it survives any of them* — a slogan, not a new theorem, where “survives” means “is determined by the ε -neighbourhood” exactly as quantified by Theorem 1 (worst case) and Theorem 3 (almost-everywhere rate). The representation changes which operations are cheap (comparison for digits, mediants for the tree, $SL_2(\mathbb{Z})$ -action for geodesics), not what is knowable.

6. Noise, lattices, and pooling

Real observations are corrupted beyond rounding. With accuracy $|x_{\text{obs}} - p/q| \leq \delta$, consider the planar lattice generated by $(1, x_{\text{obs}}/\delta)$ and $(0, 1/\delta)$. It contains $v = (q, (qx_{\text{obs}} - p)/\delta)$ with $\|v\| \leq \sqrt{2}Q$, while the Gaussian heuristic puts the shortest *irrelevant* vector near $\sqrt{\det}/\text{const} \sim \delta^{-1/2}$. Heuristically, then, lattice reduction finds the truth when

$$Q^2 \delta \lesssim 1,$$

which at $\delta = 10^{-k}/2$ is Theorem 1 again — but the lattice formulation *degrades continuously* in δ where interval bisection is brittle, and it generalises:

Pooling (heuristic + verified). For n decimals $x_1, \dots, x_n$ sharing one denominator (components of a probability vector, entries of a rational matrix row), the $(n+1)$ -dimensional analogue has determinant δ^{-n} and the same Gaussian-heuristic argument predicts success when

$$Q^{n+1} \delta^n \lesssim 1.$$

Converting that into digits needs care, because the natural reading overstates it. With $\delta = 10^{-k}$ the condition reads $(n+1) \log_{10} Q \leq nk$, while a *single* observation of K digits requires $2 \log_{10} Q \leq K$. Equating the two gives the effective precision of pooling:

$$K_{\text{eff}} = \frac{2n}{n+1} k \xrightarrow{n \rightarrow \infty} 2k,$$

not nk . Pooling is subject to diminishing returns: it buys at most a doubling of effective precision no matter how many components share the denominator, because each new component adds one lattice dimension as well as k digits of evidence. For four 3-digit observations that is $K_{\text{eff}} = 4.8$ digits — comfortably enough to pin $q = 61$ (which needs $2 \log_{10} 61 \approx 3.6$), and a good illustration that a modest constant-factor gain can still flip a problem from ambiguous to identifiable. These success conditions are heuristics, not theorems — the Gaussian heuristic can fail for structured lattices — but the implementation checks every extracted candidate against the exact consistency bound $|qx_i - p_i| \leq q\delta_i$ and, when no extra noise is declared, additionally verifies that the observed digit strings are regenerated exactly, so a heuristic failure costs recall, never soundness; randomized tests in the suite probe the claimed regime. Recall itself is then repaired exactly: when the lattice finds nothing at moderate bounds, `reconstruct_shared` falls back to an exact $O(Qn)$ scan over the observation cells, so “no common denominator exists with $q \leq Q$ ” is only ever asserted as a *theorem about the enumeration*, never as a lattice guess (an LLL miss at $Q = 100$ that the scan repairs: rounded $0.750, 0.293 \mapsto (3/4, 27/92)$).

The API grades its claims, and the levels are not interchangeable — each needs strictly more enumeration than the last:

1. **Verified** (LLL fast path): the returned q is consistent, but a short lattice vector need not be the shortest, so q may be non-minimal.
2. **Smallest** (`minimize=True`): scan $q = 1, 2, \dots$ and stop at the first feasible denominator. Testing the *nearest* numerator suffices here, because the nearest integer minimises $|qx_i - p_i|$: if any numerator works at q , the nearest one does.
3. **Unique** (`identify_shared`): this is where the nearest-numerator shortcut becomes unsound. A cell of width w admits $\lfloor wq \rfloor$ or so numerators, and as soon as $wq \geq 1$ two distinct hypotheses share one denominator — e.g. the cell $[0.95, 1.05)$ of the observation “1.0” contains both $1 = 20/20$ and $19/20$. Sampling one numerator per denominator therefore *undercounts the fibre and can certify uniqueness falsely*; the implementation takes the full Cartesian product of the admissible numerator ranges and counts distinct reduced tuples, so a consistent q and its multiples collapse to one hypothesis while two numerators at one q count twice. With a single observation it agrees with `identify` by construction (randomised test in the test suite). Four 3-digit decimals identify a $q = 61$ that no single 3-digit decimal can (each observation’s top-ranked individual candidate differs from the generating fraction — see `demo.py`, section 4). Implementation: exact-arithmetic LLL in `ratrecon.py`.

7. Constrained denominators and learned priors

7.1 Restricted denominator sets

Real sources rarely draw q uniformly: prices move in ticks ($q \in \{2^a\}$ or $\{100, 10000\}$), probabilities are quoted in basis points, musical and calendrical ratios favour smooth q . Restricting to an admissible set S (with divisor-closure: $q' \mid q \in S$ is admissible, since p'/q' is representable over q) changes the information budget twice over:

- **Entropy:** identifying a member of $\{p/q : q \in S\}$ costs about $\log_2 \sum_{q \in S} \varphi(q)$ bits instead of $2 \log_2 Q - 1.72$ — for sparse S , dramatically fewer digits suffice.
- **Geometry:** the separation improves, and the right invariant is the *common grid*, not the product of denominators. Restricting to $q, q' \leq Q$ within S and reusing $1/(qq')$ would give $\max\{qq'\} = Q^2$ again — no sharpening at all. But every admissible fraction with denominator dividing some $m \in S$ lies on the grid $\frac{1}{L}\mathbb{Z}$ with $L = \text{lcm}(S)$, so distinct admissible fractions satisfy

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \geq \frac{\gcd(q, q')}{qq'} = \frac{1}{\text{lcm}(q, q')},$$

with equality attained (Bézout makes $|pq' - p'q|$ range over the nonzero multiples of $\gcd(q, q')$). So the family's minimum spacing is *exactly* $1/M$ with

$$M = \max_{q, q' \in S} \text{lcm}(q, q'),$$

and k digits suffice as soon as $10^k \geq M$ — **linear** in the grid where the unconstrained bound is quadratic in Q . Two refinements matter here. First, M is the worst *pair*, not $\text{lcm}(S)$: for $S = \{6, 10, 14\}$ the total lcm is 210, but no two members reach it (the worst pair is (10, 14) with lcm 70), so two digits suffice where the cruder bound would demand three. Second, this is *sufficient* and based on the exact spacing, but not proven minimal — identifiability can survive a coarser grid when the extremal pair never shares a cell, just as individual fractions beat the unconstrained bound. Basis points ($S = \{10^4\}$) need $k = 4$ rather than the 8 that $Q = 10^4$ would demand unrestricted; dyadic ticks to $1/64$ need $k = 2$. Implemented as `minimum_spacing_denominator / sufficient_digits_for_denominators`. Enumeration changes shape too. Admissibility is divisor-closed, so the search runs over the *divisors* of the family members that are at most Q — found in $O(\sum_{m \in S} \min(\sqrt{m}, Q))$ — and then over the cell's numerators for each such denominator, replacing the Farey walk. Scanning the generators themselves would be exponentially wasteful: $S = \{10^9\}$ with $Q = 10$ admits just $\{1, 2, 4, 5, 8, 10\}$.

A sharpening of Theorem 1 falls out of the same argument even without constraints. Ambiguity at exactly the threshold width $10^{-k} = Q^{-2}$ would need two admissible fractions at the extremal gap $1/(qq') = Q^{-2}$, which forces $q = q' = Q$; but two *distinct* fractions with the same denominator Q differ by at least $1/Q \gg 1/Q^2$. So the true minimum gap is $\geq 1/(Q(Q-1))$, strictly bigger than $1/Q^2$, and

$$10^k \geq Q^2 \quad \text{already suffices.}$$

`ratrecon.sufficient_digits` implements this sharp form — `sufficient_digits(100) = 4`, not 5 — and the claim is verified exhaustively in the test suite: at $k = 4$, $Q = 100$, every one of the 3043 proper reduced fractions is recovered exactly, in both rounding and truncation modes. (The bound remains *sufficient, not necessary*: individual fractions are often identifiable with fewer digits — $0.333 \mapsto 1/3$ at $k = 3$ — but not all of them.)

7.2 Learned priors: propose, then verify

Below the threshold the ranking problem is pure prior selection (section 3), and a prior can be *learned*: train a sequence model on the source distribution and use its denominator posterior as proposals. The architecture that respects the mathematics is **proposer–verifier**: the network proposes a shortlist of denominator grids (a proposed value admits the reduced denominators dividing it); exact arithmetic keeps only consistent fractions over those grids, and ranks them. The network can raise recall but can never overrule hard evidence — a learned guess is never presented as a certificate.

Empirically (`results/`, reproducible via `hybrid_gru.py`): results are mean $\pm$ sample sd over three runs on one fixed fraction split; each run varies model initialisation and the training and evaluation corruptions, and every method is evaluated on the same corrupted rows within a run. With 12-digit inputs and 10% per-digit corruption, hybrid exact-fraction top-1 is 0.703 ± 0.007 against 0.685 ± 0.008 for the classical baseline; at denominator level the hybrid reaches 0.725 vs 0.694, and the GRU alone 0.490 ± 0.002 . The aggregate gain suggests the learned and classical proposals make some complementary errors; per-example overlap, corruption-position effects and other noise levels are not analysed in this pilot (one bound, one split, one architecture, three runs). On clean inputs the same hybrid nearly matches the classical optimum (0.997 vs 1.000), i.e. the learned proposals cost almost nothing when they are not needed. The scope of the idea is sharply bounded, and section 8 says by what: learned proposals help against **noise**, where the information is present but scrambled, not against **missing information** — below the threshold the fibre is genuinely many-to-one, and no proposer, learned or otherwise, can beat its limits.

8. What is inevitably lost

Four distinct mechanisms, in decreasing order of severity:

1. **Cardinality.** Below threshold, ρ_k is $\sim \frac{3}{\pi^2} Q^2 10^{-k}$ -to-one; no algorithm recovers what the fibre does not determine. Only priors (§3) or extra structure (§6) help.
2. **Discontinuity.** The denominator map $p/q \mapsto q$ is unbounded on every interval, so no *continuous* function of the observed value can compute it globally, and an exact reconstructor over unbounded height is discontinuous everywhere. Bounded to $q \leq Q$ with finite digit strings the map becomes a finite classification problem a network can in principle represent — the operative difficulty is generalisation to unseen digit sequences, not topological impossibility (the framing for the experiment in `hybrid_gru.py`).
3. **Certification gaps.** Determining p/q needs $O(\log q)$ digits, but even *witnessing* its period twice — the evidence bar `periodic_candidates` demands before proposing — needs $s + 2t \leq k$ with $t = \text{ord}_{\tilde{q}}(10)$ as large as $q - 1$ (full-reptend primes 7, 17, 19, 23, 29, 47, 59, 61, 97, ...): exponentially more digits than the value itself requires. And no finite window certifies that the unseen continuation stays periodic at all, which is why proposals are

verified against the window and never promoted to certificates.

4. **Representation friction.** Even above threshold, converting between coordinatisations is not free: the Lochs constant $0.9702\dots < 1$ is a fixed conversion loss between decimal and continued-fraction information, with a.e.-vs-worst-case gaps on top (§4).

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